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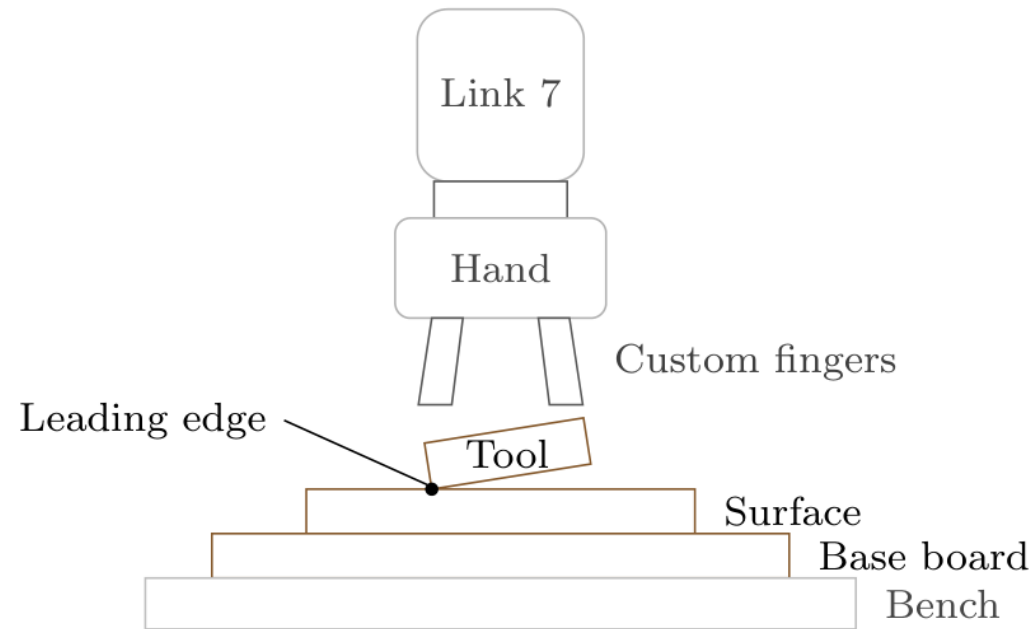
Design and Implementation of a Real-Time Impedance Controller for a 7-DOF Collaborative Robot

Heykel Khadhraoui

Computational Engineering (M.Sc.)

Supervisors: Prof. Norbert Nitzsche · Prof. Ruth Otto

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- Problem

- Tilted tool → edge-first contact
- Rigid tracking resists corrective rotation

- Main thesis question

- **How do Cartesian impedance settings affect passive alignment?**

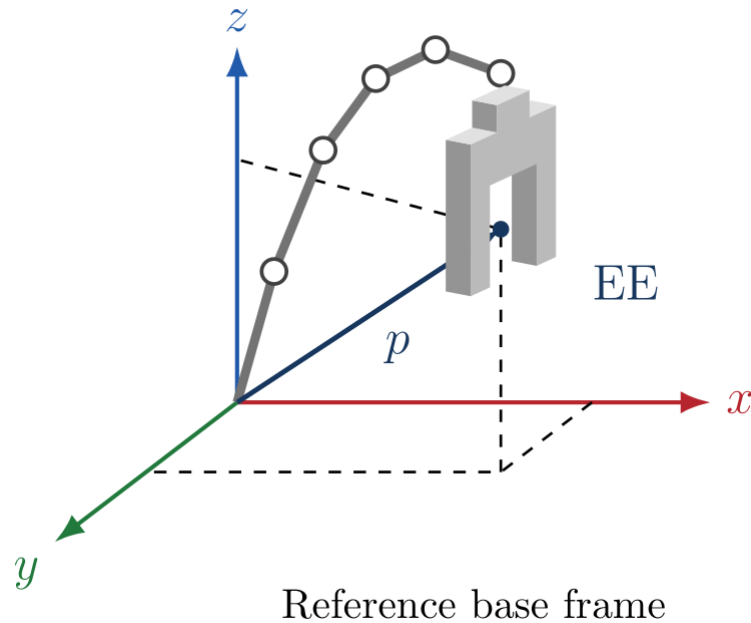
- Idea

- Normal pressing with contact-induced rotation

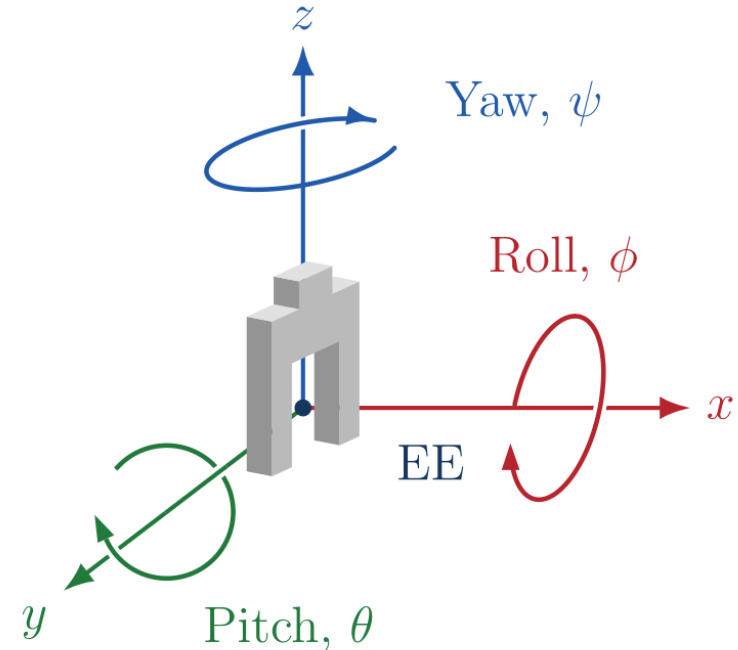
1. Introduction
2. Controller design
3. Experimental methodology
4. Validation
5. Experimental results
6. Conclusion and future work

In 3D, the end effector's position and orientation have 6 degrees of freedom.

- Position: 3 translations



- Orientation: 3 rotations



- The EE position depends on the configuration of all 7 joint angles: $p = p(q)$

- Position correction by force

$$e_p = p_d - p_{EE}$$

$$f = K_p e_p + D_p (\dot{p}_d - \dot{p}_{EE})$$

- Orientation correction by moment

$$e_R = \phi u, \quad R_d R_{EE}^\top = R(u, \phi)$$

$$m = K_R e_R + D_R (\omega_d - \omega_{EE})$$

- Combined Cartesian wrench

$$F = \begin{bmatrix} f \\ m \end{bmatrix} = \underbrace{\begin{bmatrix} K_p & \mathbf{0} \\ \mathbf{0} & K_R \end{bmatrix}}_{K_c} \begin{bmatrix} e_p \\ e_R \end{bmatrix} + \underbrace{\begin{bmatrix} D_p & \mathbf{0} \\ \mathbf{0} & D_R \end{bmatrix}}_{D_c} \begin{bmatrix} \dot{p}_d - \dot{p}_{EE} \\ \omega_d - \omega_{EE} \end{bmatrix}$$

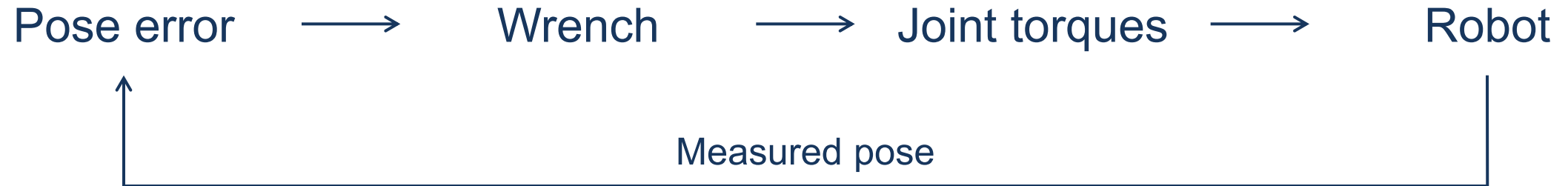
- Zero off-diagonal blocks: translation and rotation are decoupled.

K_p, K_R translational / rotational stiffness

D_p, D_R translational / rotational damping

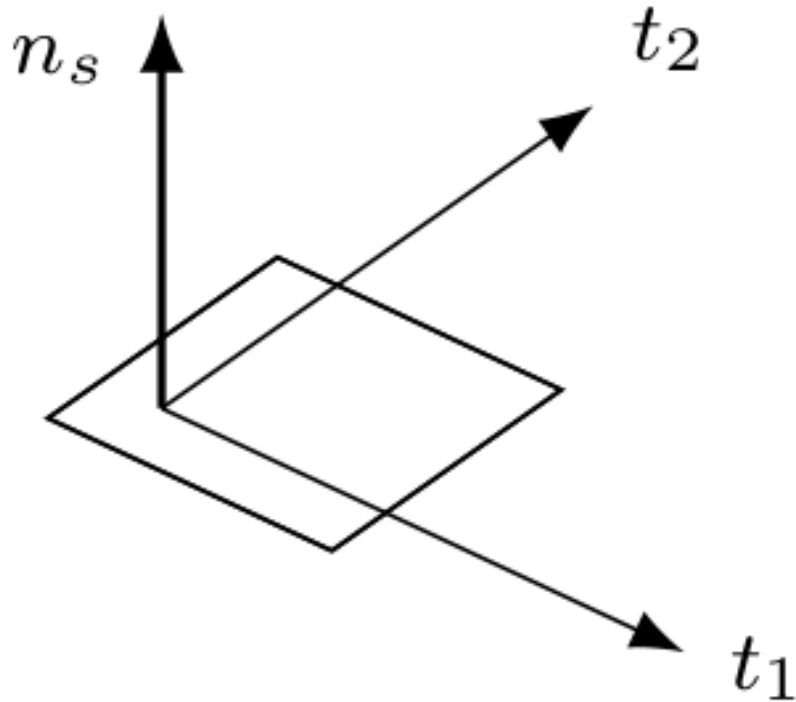
$\dot{p}, \omega$ linear / angular velocity

d, EE desired / measured end effector



$$F = \begin{bmatrix} f \\ m \end{bmatrix}, \quad \tau_{\text{cart}} = J^{\top}(q)F$$

1 kHz control loop · C++ / libfranka



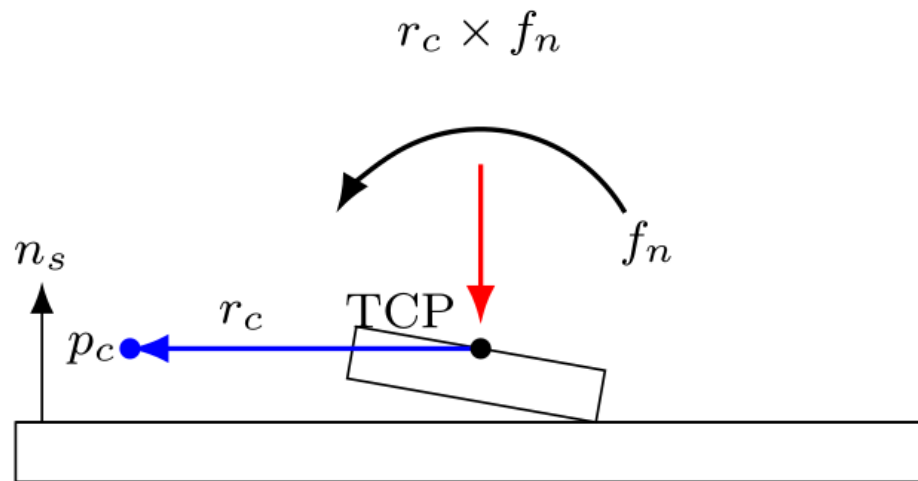
- n_s outward surface normal
- t_1, t_2 two perpendicular surface tangents
- Translation
 - Softer normal pressing
 - Stiffer motion along the surface
- Rotation
 - Softer rotations about the tangents
 - Stiffer rotation about the normal

Centre of compliance (CoC)

- Virtual reference point

p_c CoC position

r_c Displacement from TCP to CoC

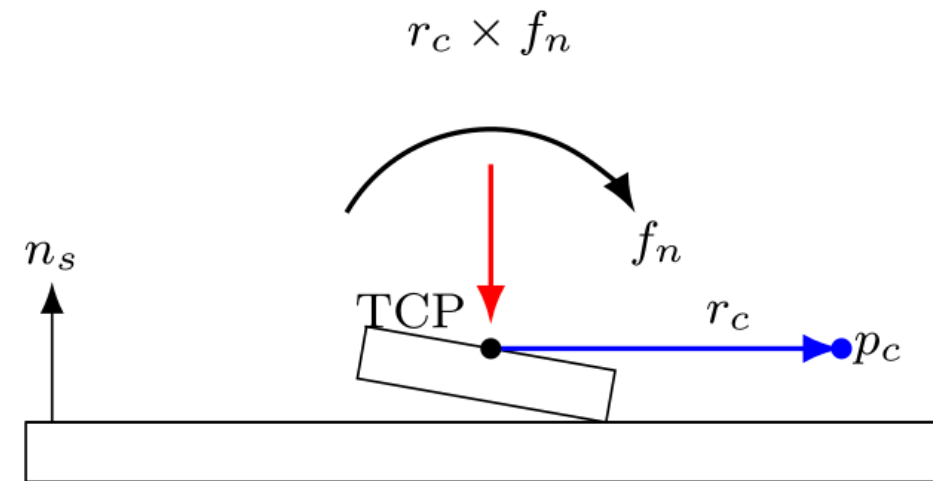


(a) Supporting displacement

- Commanded TCP moment

$$m = m_R + r_c \times f$$

m_R rotational spring-damper moment



(b) Reversed displacement

- Opposite displacement reverses the added moment

- Shift from TCP to centre of compliance

$$A = \text{Ad}(r_c), \quad e_c = Ae$$

Local small-angle model with uncoupled stiffness and damping at the CoC

- Stiffness

$$K_{\text{TCP}} = A^\top K_c A$$

- Damping

$$D_{\text{TCP}} = A^\top D_c A$$

- A non-zero CoC displacement creates translation–rotation coupling.
- Translational error and velocity can contribute to commanded moment.

- Redundancy

7 joints for a 6-DOF Cartesian pose.
One redundant direction at full rank.

$$\dim(\ker J) = 7 - 6 = 1$$

- Cartesian task preservation

Zero instantaneous TCP translation and rotation

$$J(q) \dot{q}_{\text{null}} = 0$$

- Damping τ_d

Opposes redundant joint velocity.
Suppresses unwanted joint motion.

- Conditioning τ_σ

Seeks better Jacobian conditioning.
Can initiate joint motion.

- Singularity: a configuration where the Jacobian loses rank.

Tool orientation → Surface approach → Contact establishment → Surface grinding

- Compliant contact settings

- Normal translation

$$K_{p,n} = 350 \text{ N/m}$$

- Rotations about the surface tangents

$$K_{R,t_1} = K_{R,t_2} = 5 \text{ N m/rad}$$

- Tangential translation

$$K_{p,t_1} = K_{p,t_2} = 2000 \text{ N/m}$$

$$\zeta = 1.0 \text{ inertia-scaled damping}$$

- Controlled comparisons

- A: TCP baseline for zero, positive and negative entry tilt

- B: rotational stiffness about t_1

- C: translational stiffness along t_2

- D: tangential CoC displacement

19 settings × 3 trials = 57 contact trials

- Planar contact with a 40 × 120 mm tool and a fixed entry reference for 5 s

Null-space experiment

- Commanded joint disturbance

$$\tau_{\text{dist}} = J_p(q)^\top f_{\text{dist}}$$

Joint-torque equivalent of a virtual 20 N point force on link 3.

- Cartesian pose hold

$$\|p_d - p_{\text{EE}}\| < 2 \text{ mm}$$

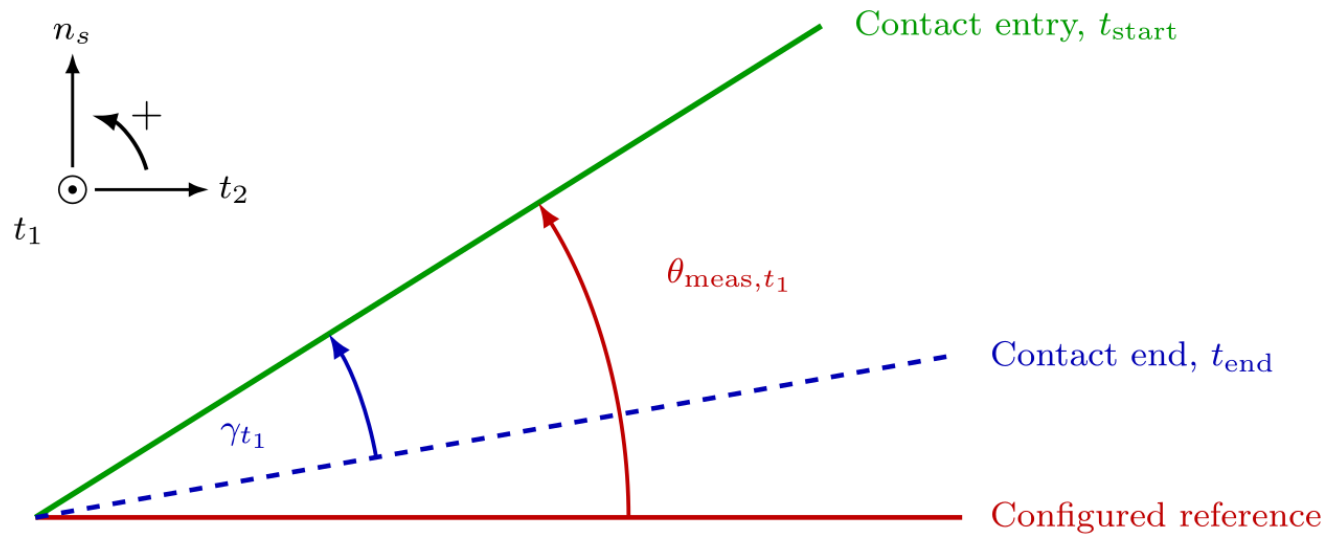
Hold the captured Cartesian pose. Assess TCP position retention.

- Four settings with three trials each

- No null-space torque
- Damping: 2 N m s/rad

- Conditioning: 1.5 and 2.0 N m
- Compare accumulated motion and net displacement

- The disturbance excites joint motion while the Cartesian controller holds the TCP.

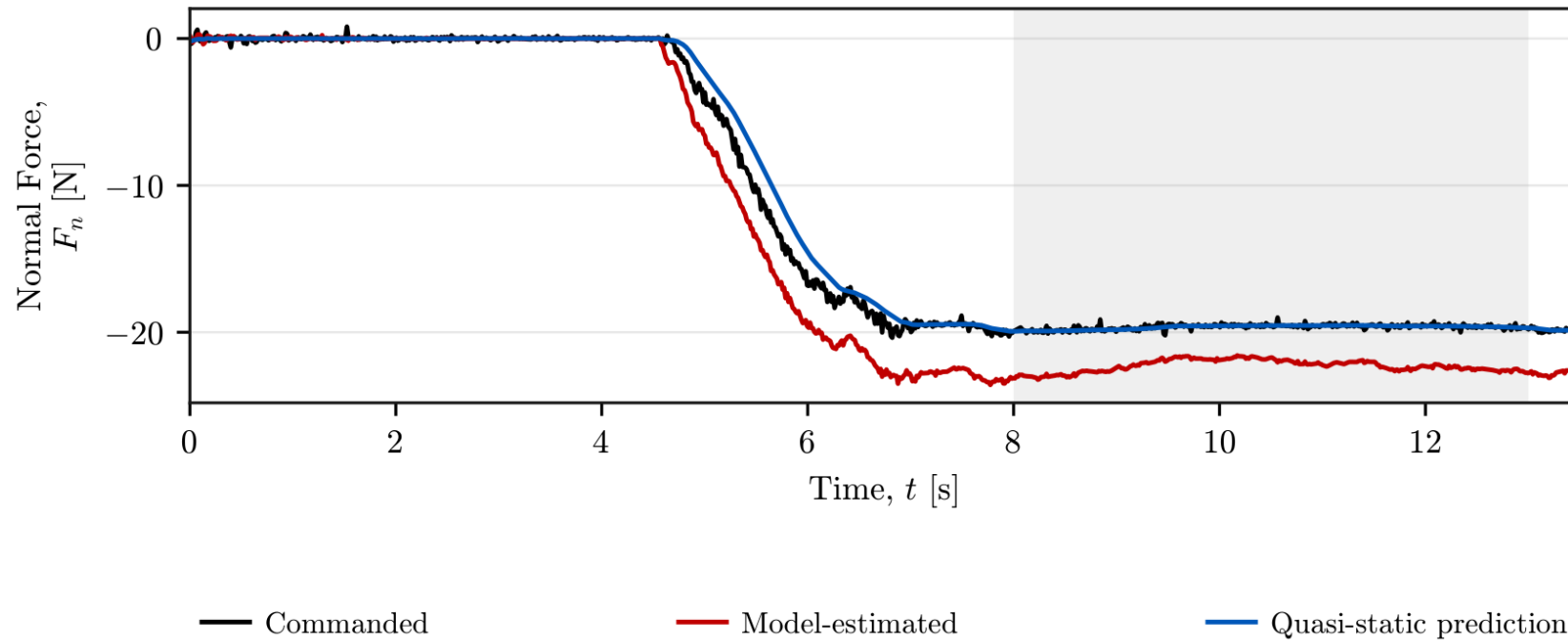


θ_{meas, t_1} Measured angular offset
Relative to the configured
surface reference

γ_{t_1} Contact response
Rotation from contact end
back to entry orientation

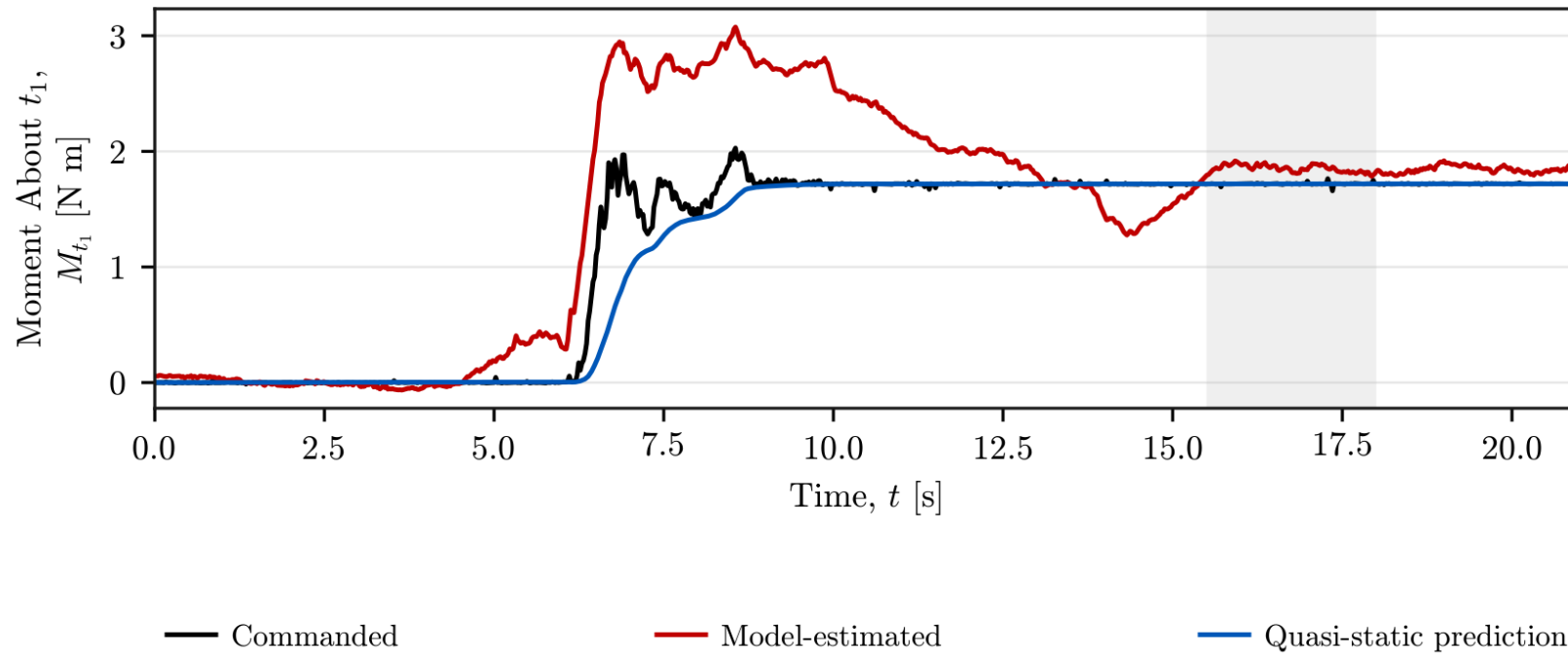
- Aligning response: same sign as the measured angular offset
- Calculated from measured end-effector orientation

$$F_{n,qs} = K_{p,n} n_s^\top (e_p - e_{p,0})$$



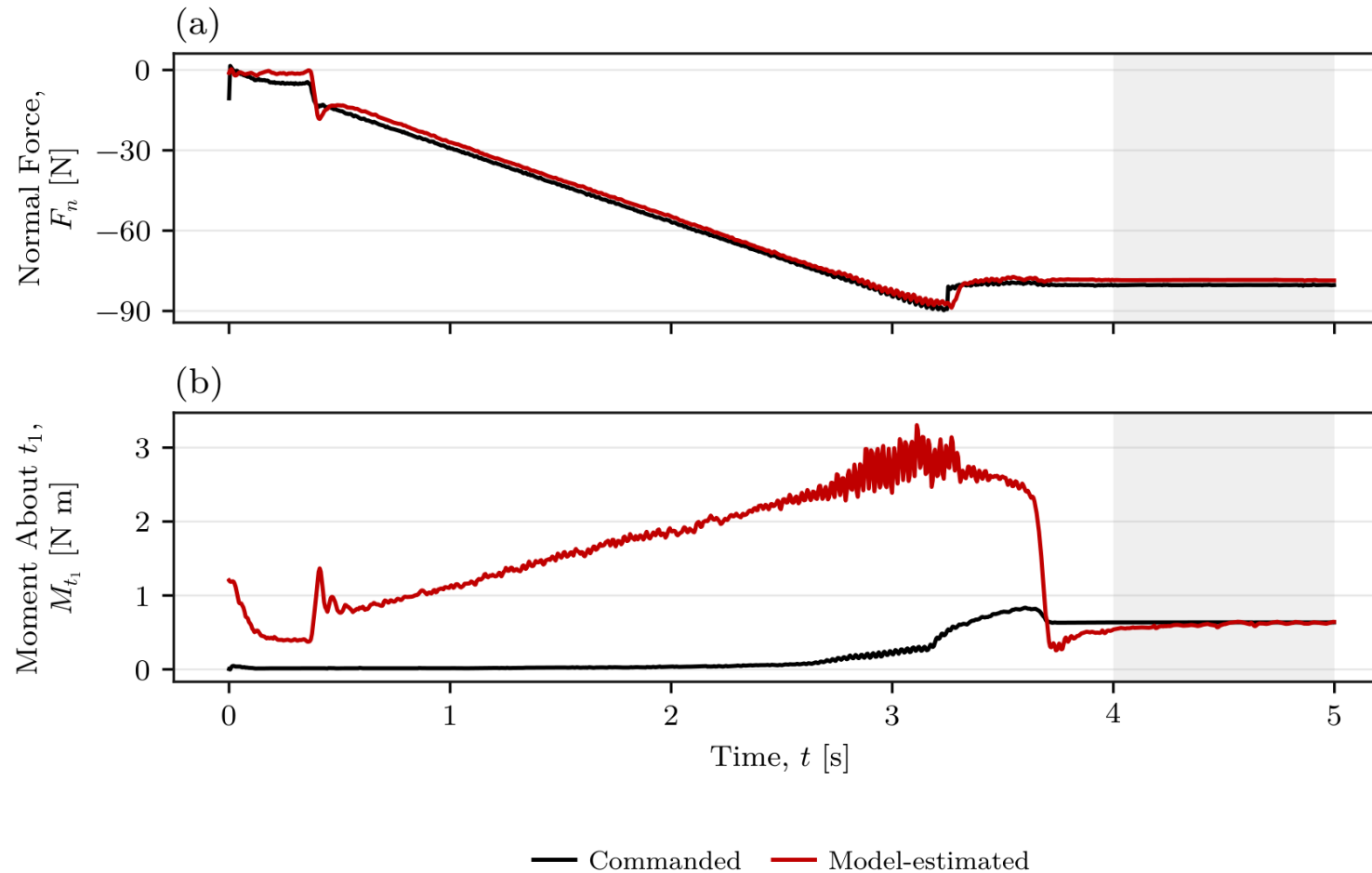
- Spring vs command: 0.05% difference
- Estimate vs command: 13.50% difference

$$M_{t_1,qs} = K_{R,t_1} t_1^\top (e_R - e_{R,0})$$



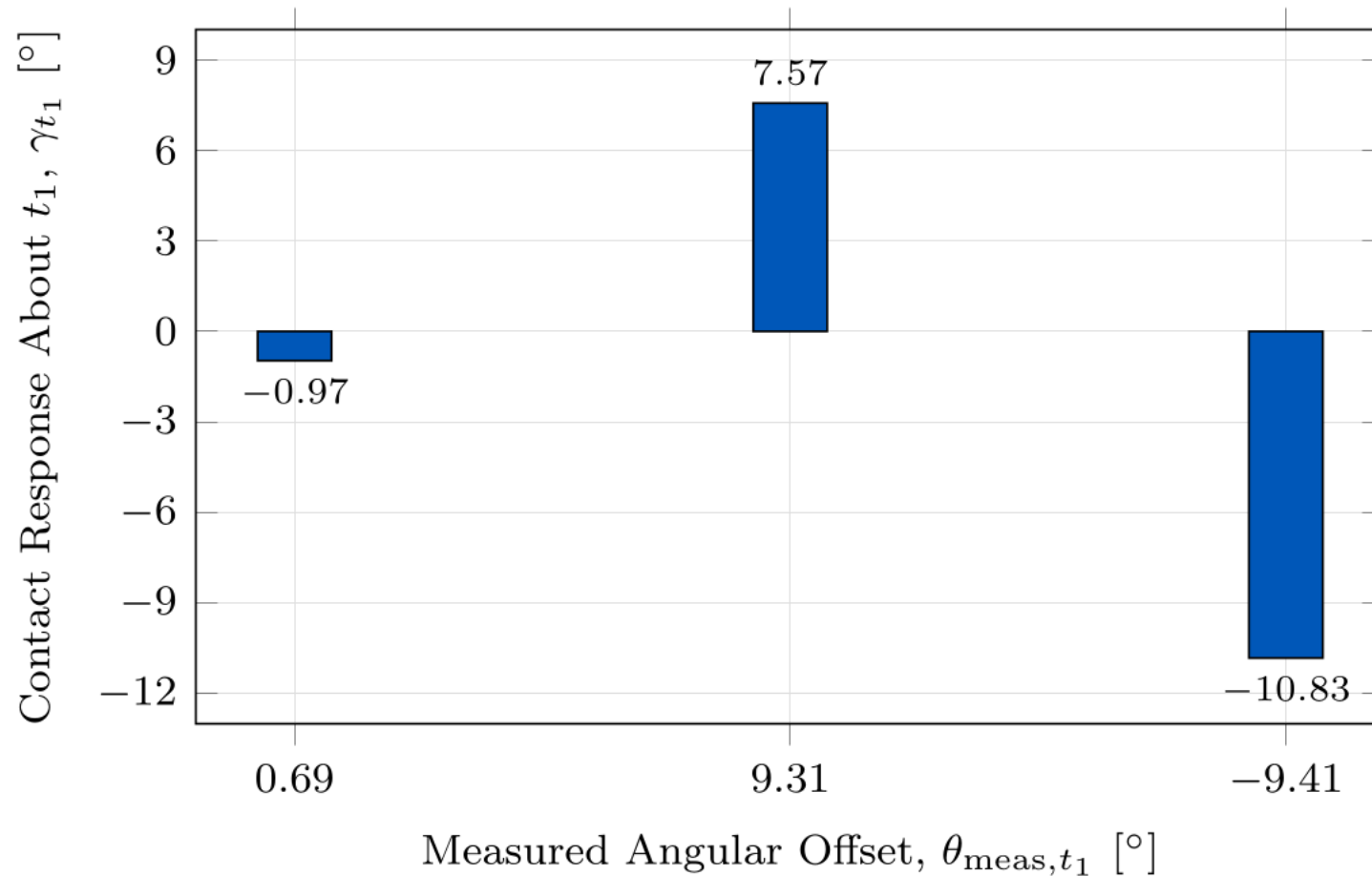
- Spring and commanded mean: 1.719 N m
- Model-estimated mean: 1.852 N m

Commanded and estimated wrench



- Stationary force difference: 2.32%
- Stationary moment difference: 4.87%
- TCP-centred trial with common axes and reference point

Baseline contact response

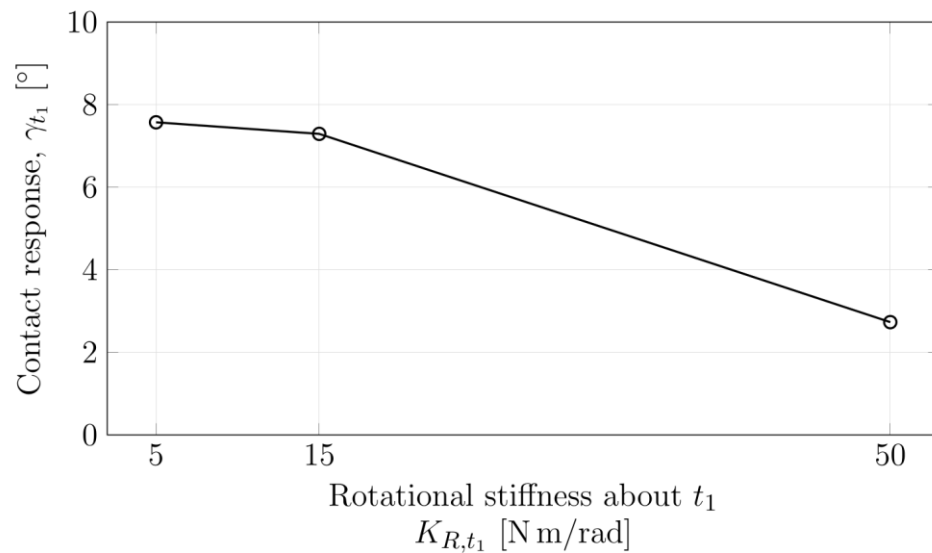


- Centre of compliance at the TCP
- Alignment response for both tilt signs
- Unequal response magnitudes

Effect of stiffness

- Rotational stiffness

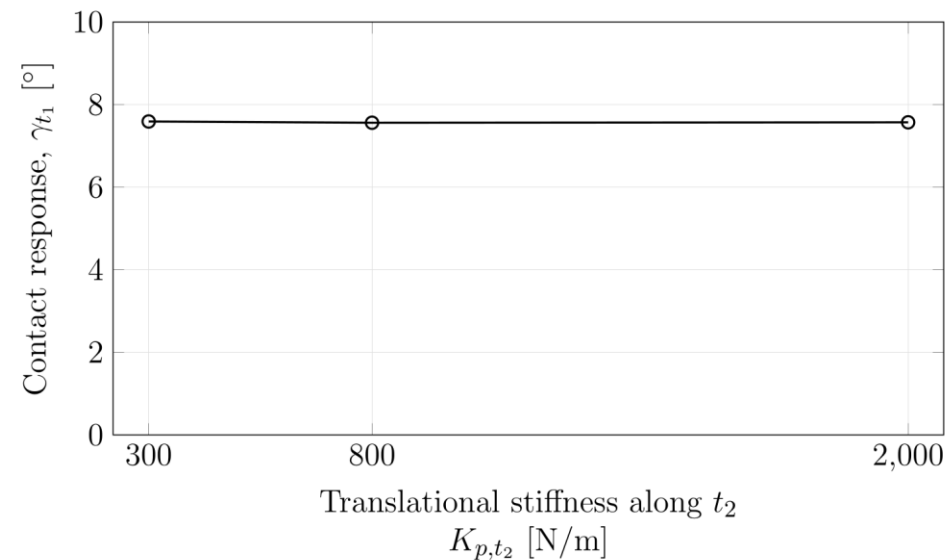
Resistance to rotation about t_1



—○— Measured angular offset
 $\theta_{\text{meas},t_1} = (9.31\text{--}9.32)^\circ$

- Translational stiffness

Resistance to sliding along t_2

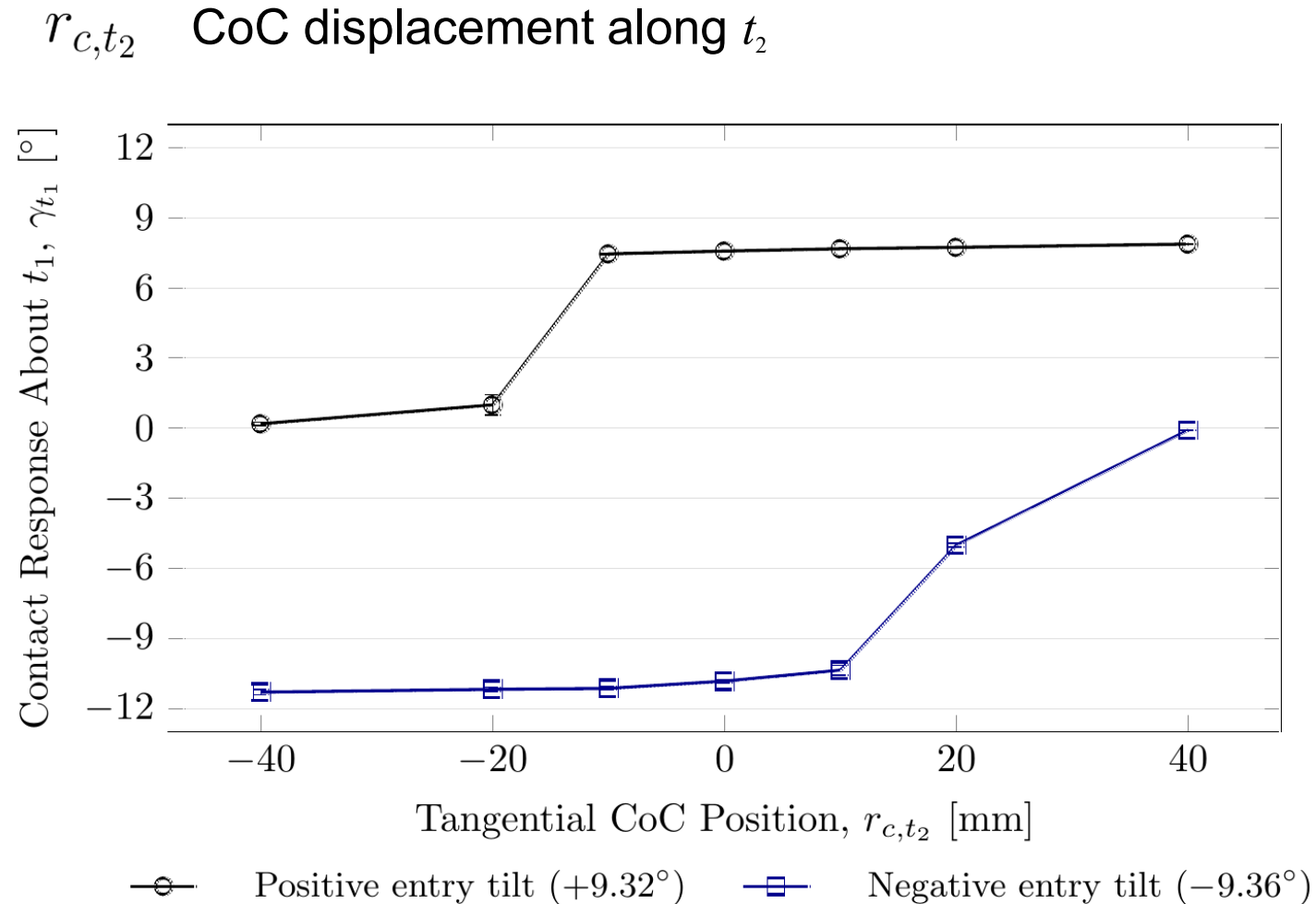


—○— Measured angular offset
 $\theta_{\text{meas},t_1} = (9.24\text{--}9.31)^\circ$

- Rotational stiffness: 63.9% lower response

- Tangential stiffness: 0.4% response span

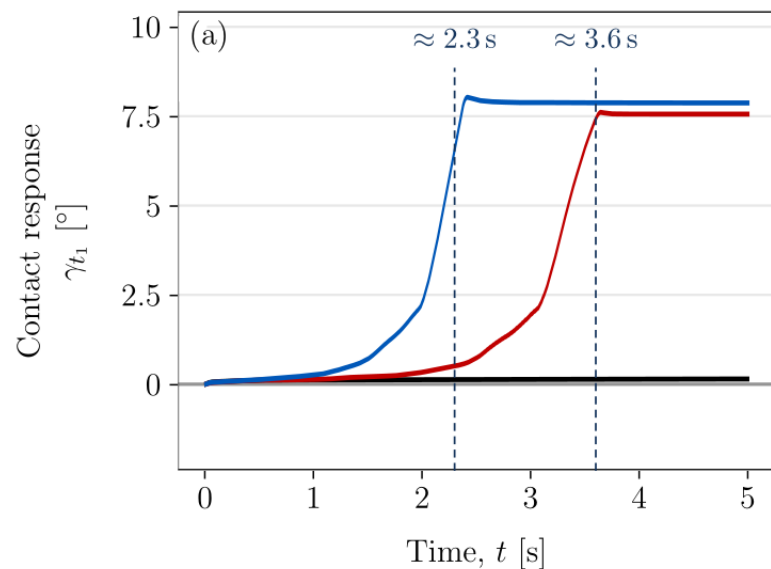
Effect of CoC position



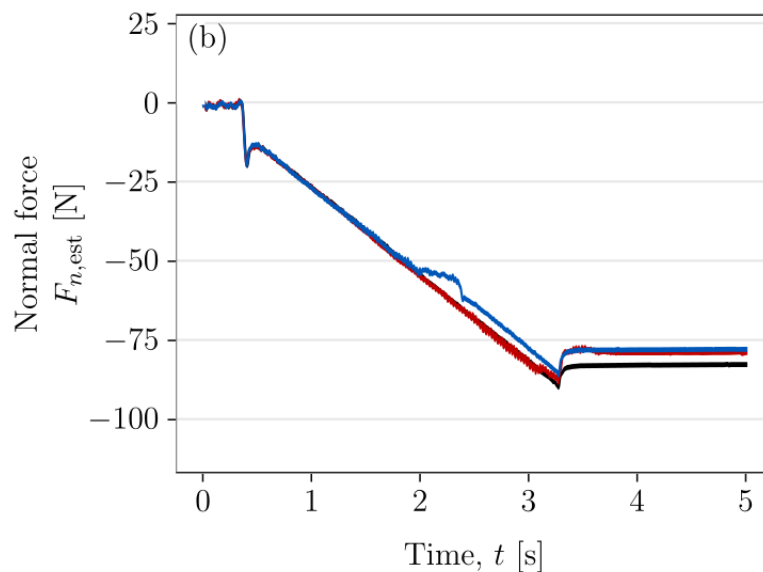
Relative to the TCP-centred baseline

- Supporting displacement: about 4% higher response
- Opposite side: 98–99% lower response
- Required side reverses with the entry tilt

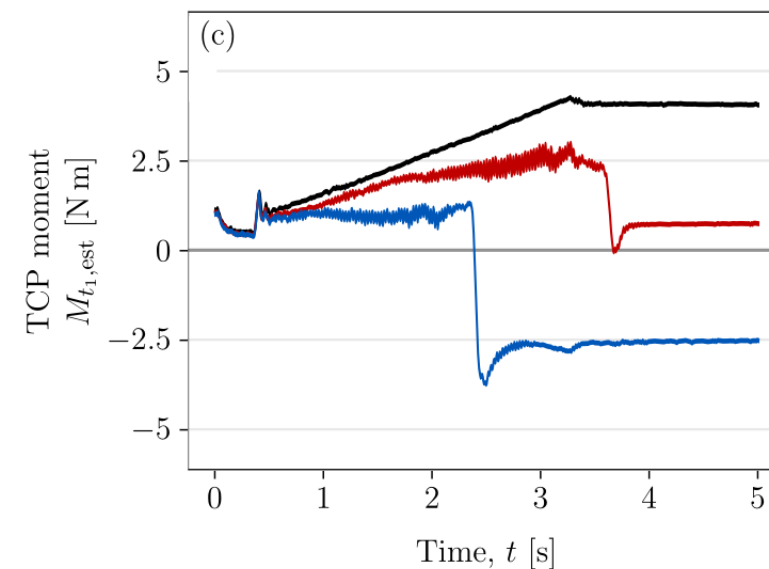
- Contact response



- Normal force

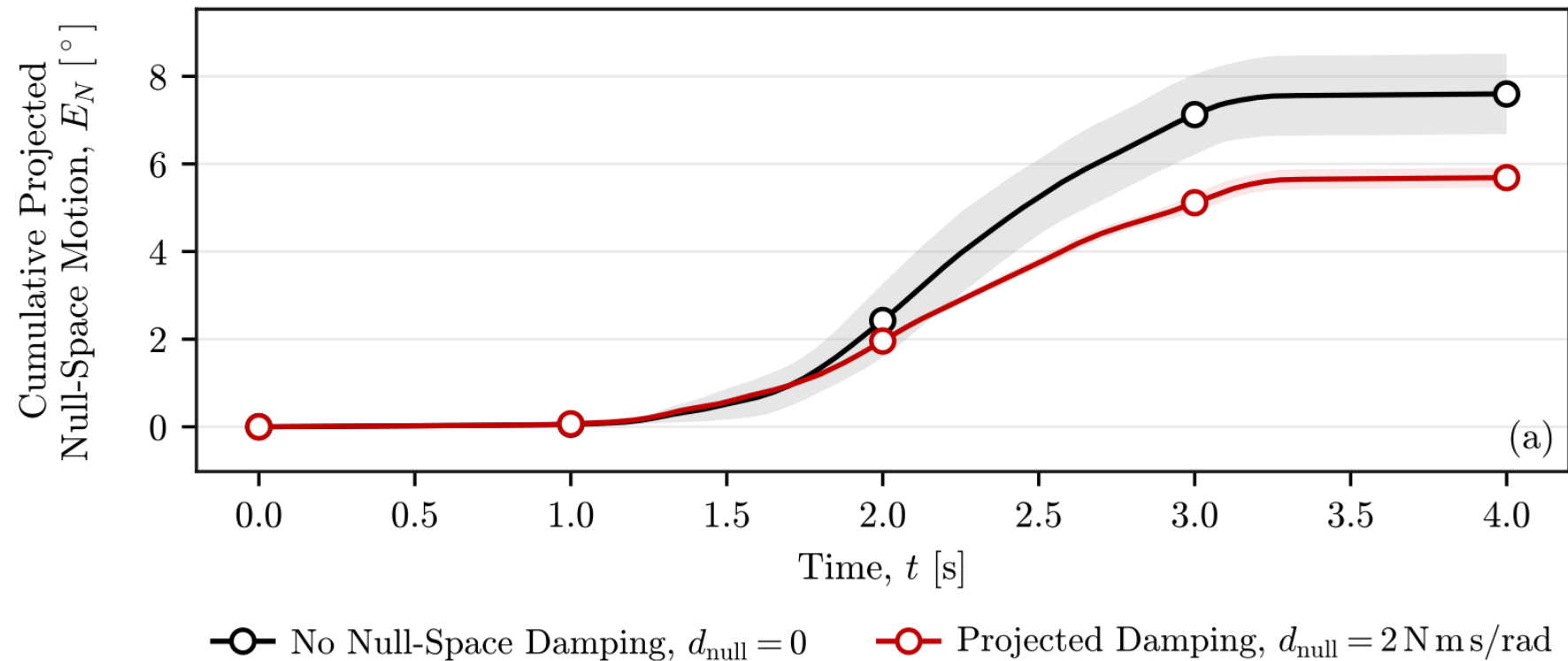


- TCP moment

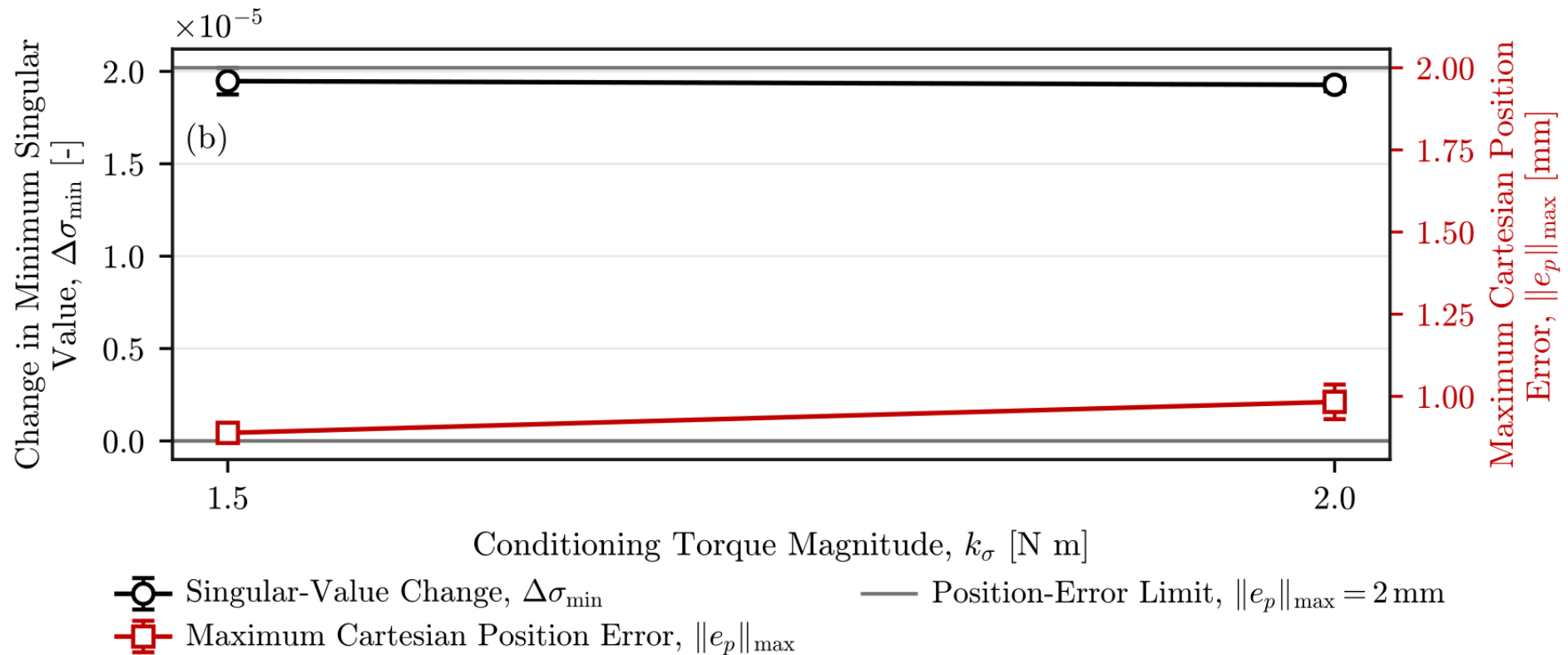


- Reversing CoC displacement reverses the interaction moment
- Supporting displacement: about 1.3 s earlier in these representative traces
- Final normal force (approx.): -83 N (-40 mm), -79 N (TCP), -78 N (+40 mm)

Null-space results

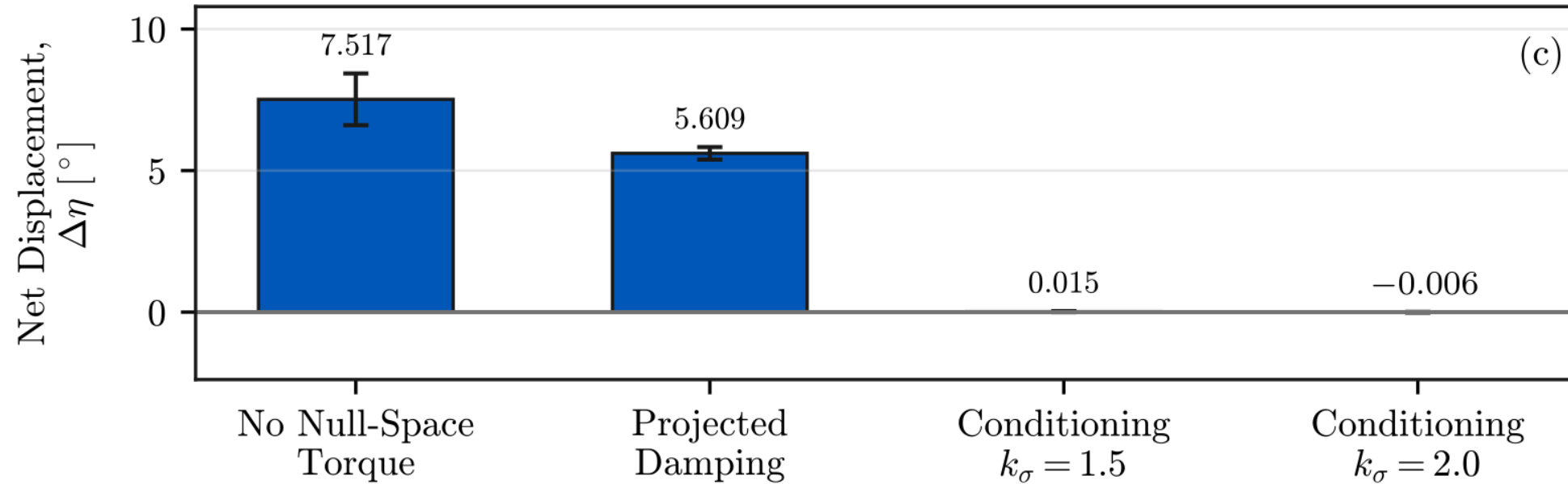


- Damping reduced cumulative motion from 7.596° to 5.687° (25%).
- Identical commanded disturbance across three trials per setting



- Both settings improved the local conditioning indicator.
- Maximum TCP error across all trials: 1.304 mm (criterion: 2 mm).

Net null-space displacement

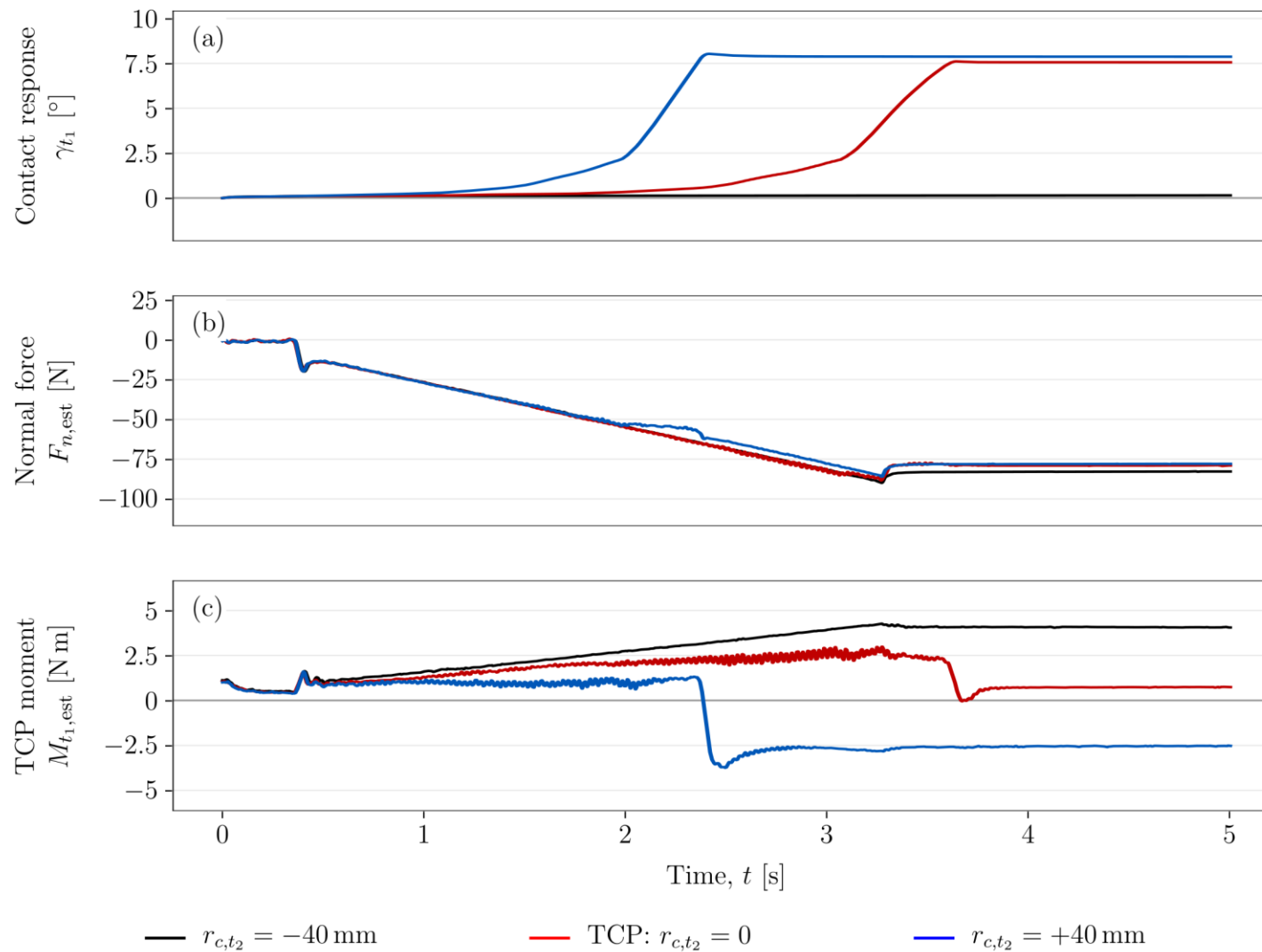


- Conditioning reduced the net displacement to nearly zero.
- Small net displacement can coexist with back-and-forth motion.

Conclusion

- Passive alignment through Cartesian impedance
- Higher rotational stiffness reduced contact rotation
- Changing tangential stiffness had almost no effect on contact rotation
- CoC displacement supports or opposes alignment
- Damping reduced motion and conditioning countered the tested disturbance
- Future work
 - More rigid tool mount
 - Adaptive CoC selection and return to the TCP after alignment

Contact response, normal force and moment

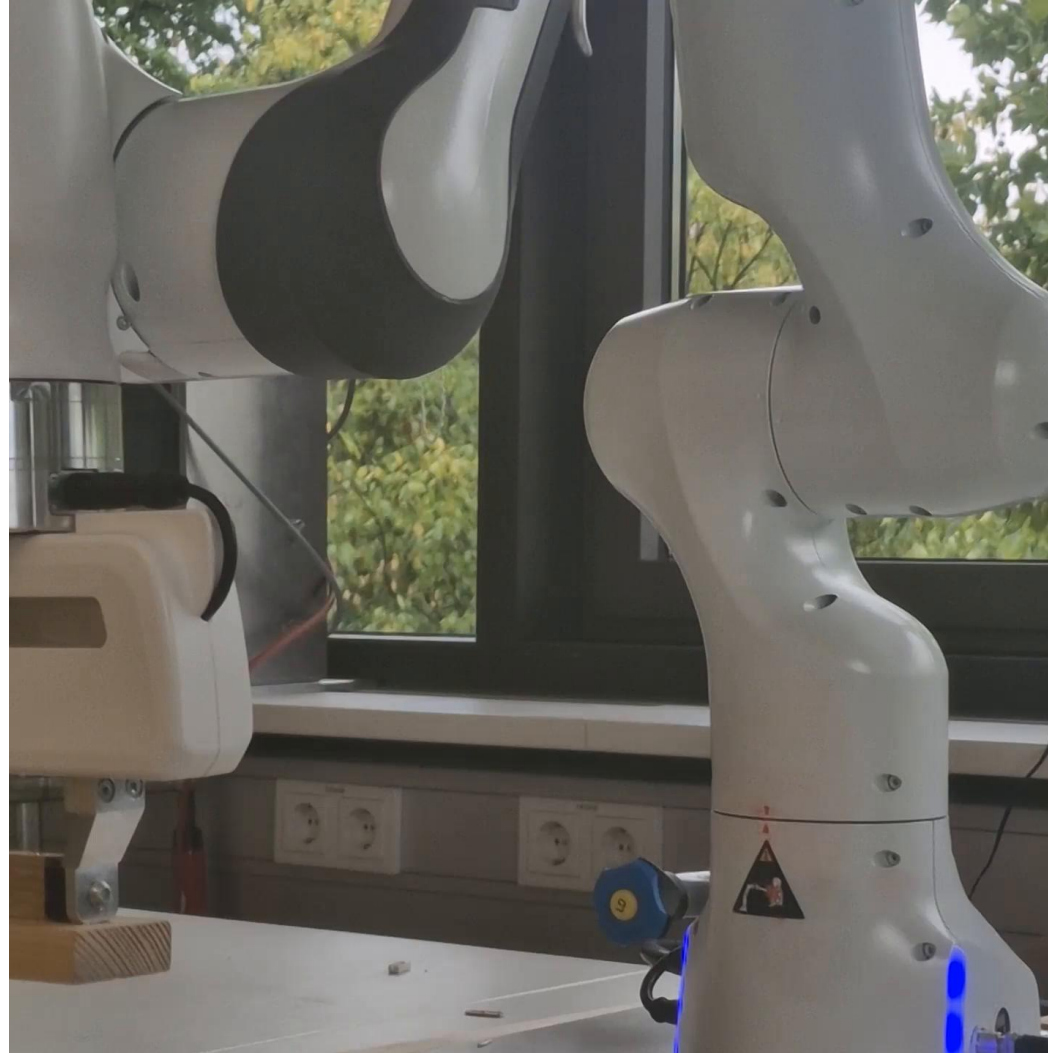


- Representative traces
Positive entry tilt
- Similar final normal force
- Moment reverses with
CoC displacement
- Force and moment:
model-based robot
estimates



Pre-grinding hold · fixed desired orientation · CoC at TCP

Disturbance demonstration



No null-space torque → Damping → Conditioning